



RESEARCH ARTICLE

Modulation of growth, yield and nutrient uptake of rice (*Oryza sativa* L.) through integrated nutrient management

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Abstract

A field experiment was carried out during kharif season of 2022–23 and 2023–24 at the Agronomy Main Research Farm, CoA, OUAT, Bhubaneswar. The experiment was laid out in randomised block design (RBD) with 12 treatments and replicated thrice. The rice variety “Kalinga Dhan 1201” was transplanted in kharif season with 12 treatments. The pooled analysis results revealed that application of T₉ resulted in maximum plant height (118.4 cm), tillers m⁻² (372), dry matter accumulation (1207 g m⁻²) at harvest. The maximum leaf area index (4.91), chlorophyll index readings (53.8) and green seeker readings (0.82) were recorded at 75 days of transplanting. Similarly, the superior yield attributing characters like panicles m⁻² (377), filled grains panicle⁻¹ (160.6), test weight (24 g) and yield parameters like grain yield (5804 kg ha⁻¹) and straw yield (6598 kg ha⁻¹) and was statistically comparable to other zinc (Zn)-supplemented treatments. The present study clearly demonstrates that integrated nutrient management (INM) is an effective strategy for enhancing rice growth, yield and nutrient uptake under irrigated conditions. Among the treatments evaluated, the application of 75 % soil test-based nitrogen (STBN) supplemented with 25 % nitrogen through vermicompost along with zinc sulphate (ZnSO₄) proved to be the most effective, resulting in superior crop performance and improved nutrient absorption.

Keywords: farmyard manure; nano urea; rice; vermicompost; yield

Introduction

Rice (*Oryza sativa* L.) commonly known as the staple food crop because of its extensive production and consumption, is one of the most significant staple food crops in the world. Rice serves as the primary source of calories for nearly half of the global population, particularly in Asia, where it forms the cornerstone of food security. Over 20 % of human calories come from rice, which makes up a significant portion of the world's grain supply (1). Asia alone produces and consumes about 90 % of the global rice, highlighting its socio-economic and nutritional importance (2). With an expected 43–45 million hectares of land and an output of about 130 million tonnes in 2023–24, India is the second largest producer of rice after China, with an average productivity of 2.9–3.0 t ha⁻¹ (3). Many smallholding farmers depend on rice production for their livelihoods, especially in eastern states like Odisha where it is essential to employment, food security and the rural livelihood. Soil fertility and nutrient availability have a major impact on rice productivity. Important physiological processes including tillering, panicle initiation, grain filling and total yield production depend on essential nutrients like nitrogen (N), phosphorus (P) and potassium

(K). However, over time, intensive rice farming has resulted in the overuse and unbalanced use of chemical fertilisers, especially nitrogenous fertilisers, which has deteriorated soil health and decreased nutrient usage efficiency (NUE) (4). Yield levels have stagnated because of continuous cropping without sufficient organic matter replenishment, which has negatively impacted the physical, chemical and biological characteristics of the soil. Additionally, small and marginal farmers can no longer afford to only rely on inorganic fertilisers due to growing fertiliser costs and decreased subsidy assistance. Conversely, sole reliance on organic nutrient sources may not meet the nutrient demands of high-yielding rice varieties due to their low nutrient concentration and slow nutrient release (5). In this regard, maintaining soil fertility and increasing agricultural yield with integrated nutrient management (INM) has become a practical and sustainable strategy. The INM requires utilising chemical fertilisers economically and in combination with organic sources including green manures, vermicompost, farmyard manure (FYM) and biofertilisers. Agro techniques to maintain rice yield, improve fertiliser use efficiency and restore soil fertility under intensive cropping include balanced

fertiliser application by promoting INM by balancing application of organic nutrient inputs (FYM, compost, vermicompost, green manures and biofertilisers) with chemical fertilisers (6). This approach helps in improving nutrient use efficiency, synchronising nutrient release with crop demand and enhancing soil organic carbon and microbial activity (7). The integrated use of organic and inorganic nutrient sources not only ensures higher and stable yields but also contributes to long-term sustainability of rice-based cropping systems.

Materials and Methods

The experiment was conducted during the kharif seasons of 2022–23 and 2023–24 at the Agronomy Main Research Farm, College of Agriculture, Odisha University of Agriculture and Technology, Bhubaneswar. It is located approximately 64 km away from the Bay of Bengal in the East and South-Eastern Coastal plain agro-climatic zone of Odisha, classified under the East Coastal Plains and Hills zone of the humid tropics of India as per the National Agricultural Research Project classification (AZ 63). A total of 12 treatments comprising different INM practices in rice were evaluated in a randomised block design (RBD) with 3 replications. The treatments are T₁- 100 % soil test-based nutrient recommendation (STBN), T₂- 75 % STBN + 25 % N through FYM, T₃- 75 % STBN + 25 % N through vermicompost (VC), T₄- 75 % STBN + nano urea (NU) spray, T₅- 50 % STBN + 25 % N through FYM + NU spray, T₆- 50 % STBN + 25 % N through VC + NU spray, T₇- 100 % STBN + zinc sulphate (ZnSO₄), T₈- 75 % STBN + 25 % N through FYM + ZnSO₄, T₉- 75 % STBN + 25 % N through VC + ZnSO₄, T₁₀- 75 % STBN + NU spray + ZnSO₄, T₁₁- 50 % STBN + 25 % N through FYM + NU spray + ZnSO₄, T₁₂- 50 % STBN + 25 % N through VC + NU spray + ZnSO₄. The experimental soil (0–15 cm depth) was acidic (pH 5.04), non-saline (EC 0.02 dS m⁻¹), with a bulk density of 1.43 Mg m⁻³ and organic carbon content of 5.23 g kg⁻¹. It contained 213.3 kg N, 16.3 kg P, 134.4 kg K, 12.9 kg S ha⁻¹ and 0.52 mg Zn kg⁻¹, indicating low available N, phosphorus (P), sulphur (S) and zinc (Zn) and medium available potassium (K). The FYM and VC partially replaced inorganic fertilisers to increase soil fertility and nutrient usage efficiency, while STBN supplied the suggested nutrient demand based on soil test results. The ZnSO₄ was used to promote nutrient absorption and repair Zn shortage, while NU was used to increase N usage efficiency by replenishing crop N requirement throughout crucial growth phases. Based on suggested agronomic techniques and results from earlier research, these ratios were selected. The rice variety 'Kalinga Dhan 1201' was used as the test crop. It is a high-yielding, moderately disease-resistant variety suitable for irrigated upland conditions, with a maturity duration of 110–112 days, fine long slender grains and an average yield potential of about 5.72 t ha⁻¹. Seedlings were raised in a nursery and transplanted in the main field at a spacing of 15 cm × 10 cm, maintaining 2–3 seedlings per hill. Fertiliser application was based on soil test values and treatment specifications. Although the recommended dose of fertilizer (RDF) for rice was 80–40–40 kg N–P₂O₅–K₂O ha⁻¹, N was increased to 100 kg ha⁻¹ (100–40–40 kg N–P₂O₅–K₂O ha⁻¹) due to low available soil N. Urea, diammonium phosphate (DAP), muriate of potash (MOP) and ZnSO₄ served as inorganic nutrient sources, while farmyard manure FYM contains 0.5 % N, 0.2 % P₂O₅ and 0.5 % K₂O and VC (VC 1.5 % N, 0.3 % P₂O₅ and 0.5 % K₂O) were used as organic sources and incorporated into the soil as basal application before transplanting. Nano urea (4 % N) was applied as a

foliar spray at 21 and 40 days after transplanting (DAT) using 500 L of water per hectare. Growth and yield data were recorded at harvest from five randomly selected plants per plot and averaged for statistical analysis. Growth parameters included plant height and dry matter accumulation at harvest, along with leaf area index (LAI), chlorophyll index and Green Seeker readings at 75 DAT. Yield attributes recorded were grain yield per plant, panicle length and panicle dry weight. Growth observations were recorded from five randomly selected plants per plot. Plant height was measured periodically from 30 DAT at 15-day intervals until maturity. Tiller number was counted at similar intervals and expressed per square meter. The LAI was determined using a leaf area meter by relating leaf area to ground area. Dry matter accumulation was assessed by oven-drying plant samples at 60 ± 5 °C and expressed in g m⁻². Chlorophyll content was estimated using a SPAD meter, while crop vigour and nitrogen status were assessed through NDVI readings using a Green Seeker sensor at 75 DAT. Yield attributes included number of effective panicles, panicle length, panicle dry weight and number of filled grains per panicle. The 1000-grain weight was calculated from grain samples collected from selected hills. Grain yield was obtained from the net plot area, dried to 14 % moisture and expressed in kg ha⁻¹. Straw yield was recorded similarly after drying. Harvest index (HI) was calculated as the ratio of grain yield (economic yield) to total biological yield (grain + straw yield), expressed as a percentage following the concept proposed by Nichiporovic (8, 9). This comprehensive methodology enabled precise evaluation of INM effects on growth, yield and nutrient dynamics in rice. The experimental data were subjected to analysis of variance (ANOVA) appropriate to the experimental design. Treatment effects were tested using the *F*-test and significant differences among treatment means were compared using the Critical difference (CD) test at the 5 % level of significance.

Results

Effect of integrated nutrient management on growth attributes of rice

The plant growth parameters like plant height (118.4 cm), dry matter production (1207.2 g) and number of tillers m⁻² (372.5) at harvest was recorded with the application of 75 % STBN + 25 % VC + ZnSO₄ (T₉) to rice but was statistically at par with 75 % STBN + 25 % FYM + ZnSO₄ (T₈), 75 % STBN + NU + ZnSO₄ (T₁₀), 50 % STBN + 25 % VC + NU + ZnSO₄ (T₁₂), 50 % STBN + 25 % FYM + NU + ZnSO₄ (T₁₁) and 100 % STBN + ZnSO₄ (T₇). (1207.2 g) (Table 1). This increase in height may be attributed to application of ZnSO₄ to rice because Zn is essential for rice as it activates enzymes, promotes oxygen synthesis, improves seedling vigour, chlorophyll formation and grain filling. Zinc is an important micronutrient that limits rice growth and yield. The present findings clearly indicate that the application of integrated nutrient management (INM) practices involving 75 % STBN + 25 % vermicompost + ZnSO₄ (T₉) significantly enhanced plant growth parameters such as plant height, dry matter production and number of tillers in rice. However, the statistical parity of T₉ with other treatments (T₈, T₁₀, T₁₁, T₁₂ and T₇) suggests that a balanced combination of organic, inorganic and micronutrient sources play a crucial role in achieving optimum crop growth rather than reliance on a single nutrient source.

Table 1. Growth parameters of rice as influenced by integrated nutrient management (INM) (poolta of 2022–23 and 2023–24)

Treatments		Plant height (cm)	Tillers m ⁻²	Dry matter (g m ⁻²)	LAI	Chlorophyll index readings	Green seeker readings
		At harvest	At harvest	At harvest	75 DAT	75 DAT	75 DAT
T ₁	100 % STBN	107.2	297.5	990.3	4.49	43.0	0.61
T ₂	75 % STBN + 25 % FYM	110.7	327.0	1063.2	4.62	46.1	0.66
T ₃	75 % STBN + 25 % VC	110.8	328.8	1069.0	4.63	46.5	0.66
T ₄	75 % STBN + NU	108.4	302.5	1012.6	4.52	44.8	0.62
T ₅	50 % STBN + 25 % FYM + NU	109.5	307.5	1045.5	4.57	44.8	0.63
T ₆	50 % STBN + 25 % VC + NU	110.3	315.0	1054.7	4.62	45.3	0.63
T ₇	100 % STBN + ZnSO ₄	115.0	346.7	1120.3	4.72	51.4	0.76
T ₈	75 % STBN + 25 % FYM + ZnSO ₄	117.7	366.5	1174.8	4.85	53.4	0.81
T ₉	75 % STBN + 25 % VC + ZnSO ₄	118.4	372.5	1207.2	4.91	53.8	0.82
T ₁₀	75 % STBN + NU + ZnSO ₄	117.4	360.0	1147.2	4.80	53.0	0.80
T ₁₁	50 % STBN + 25 % FYM + NU + ZnSO ₄	115.3	348.7	1128.4	4.72	52.2	0.78
T ₁₂	50 % STBN + 25 % VC + NU + ZnSO ₄	116.8	355.5	1130.0	4.76	52.3	0.78
SEm (±)		1.56	9.60	30.49	0.066	1.07	0.020
CD (0.05)		4.4	27.4	86.9	0.19	3.1	0.06

DAT: days after transplanting; STBN: soil test-based nutrient recommendation; FYM: farmyard manure; VC: vermicompost; NU: nano urea; ZnSO₄: zinc sulphate; LAI: leaf area index; SEm (±): standard error of the mean; CD (0.05): critical difference at the 5 % probability level. RDF (recommended dose of fertilizers) of rice: 80 : 40 : 40 kg N, P₂O₅, K₂O ha⁻¹, Nano urea spray (2 sprays at 30 and 50 DAT), ZnSO₄ (25 kg ha⁻¹), phosphorus (P) and potassium (K): 40 kg each ha⁻¹ as basal.

The LAI, chlorophyll index readings and green seeker readings were recorded maximum with the application of 75 % STBN + 25 % VC + ZnSO₄(T₉) (4.91, 53.8 and 0.82, respectively) at 75 days after transplanting (DAT) but were statistically at par with 75 % STBN + 25 % FYM + ZnSO₄(T₈), 75 % STBN + NU + ZnSO₄(T₁₀), 50 % STBN + 25 % VC + NU + ZnSO₄(T₁₂), 50 % STBN + 25 % FYM + NU + ZnSO₄(T₁₁) and 100 % STBN + ZnSO₄(T₇). The LAI was maximum up to 75 DAT and thereafter declined. The chlorophyll index readings, measured with instrument SPAD (Soil-Plant Analysis Development) chlorophyll meter, increased steadily up to 75 DAT (Table 1). The maximum performed treatment T₉ has the partial substitution of inorganic nitrogen with vermicompost, in combination with Zn supplementation, improved leaf chlorophyll content and N status.

Effect of integrated nutrient management on yield attributes and yield of rice

The yield attributing characters like number of panicles m⁻² of (377) and filled grains panicle⁻¹ (160.6) resulted maximum when applied with 75 % STBN + 25 % VC + ZnSO₄(T₉) to rice crop but was statistically at par with 75 % STBN + 25 % FYM + ZnSO₄(T₈), 75 % STBN + NU + ZnSO₄(T₁₀), 50 % STBN + 25 % VC + NU + ZnSO₄(T₁₂), 50 % STBN + 25 % FYM + NU + ZnSO₄(T₁₁) and 100 % STBN + ZnSO₄(T₇) (Table 2). A delay in the onset of flowering was observed in the zinc-treated plots. The results showed that different treatments had significant impact on grain and straw yield. Zinc treated plots revealed superior grain and straw yield over other treatments. Application of 75 % STBN + 25 % VC + ZnSO₄(T₉) to rice plant

Table 2. Yield attributing characters and yield of rice as influenced by integrated nutrient management (INM) (pooled data of 2022–23 and 2023–24)

Treatments		Panicles m ⁻²	Filled grains panicle ⁻¹	Test weight (g)	Grain yield (kg ha ⁻¹)	Straw yield (kg ha ⁻¹)	Harvest index (%)
T ₁	100 % STBN	305	141.9	22.2	4544	5551	45.01
T ₂	75 % STBN + 25 % FYM	327	145.5	22.8	5015	5817	46.32
T ₃	75 % STBN + 25 % VC	328	146.3	23.0	5061	5858	46.31
T ₄	75 % STBN + NU	315	143.1	22.3	4711	5636	45.54
T ₅	50 % STBN + 25 % FYM + NU	321	143.7	22.5	4785	5651	45.86
T ₆	50 % STBN + 25 % VC + NU	325	144.8	22.7	4892	5775	45.86
T ₇	100 % STBN + ZnSO ₄	356	152.5	23.2	5493	6298	46.59
T ₈	75 % STBN + 25 % FYM + ZnSO ₄	371	157.6	23.8	5746	6517	46.86
T ₉	75 % STBN + 25 % VC + ZnSO ₄	377	160.6	24.0	5804	6598	46.66
T ₁₀	75 % STBN + NU + ZnSO ₄	370	154.2	23.7	5638	6461	46.60
T ₁₁	50 % STBN + 25 % FYM + NU + ZnSO ₄	357	152.9	23.3	5530	6327	46.64
T ₁₂	50 % STBN + 25 % VC + NU + ZnSO ₄	367	153.6	23.5	5564	6405	46.49
SEm (±)		10.29	2.83	0.75	156.69	172.92	1.57
CD (0.05)		29.3	8.1	NS	446.6	492.9	NS

Values are means of 3 replications. Pooled data represent the average of 2022–23 and 2023–24. SEm (±): standard error of the mean; CD (0.05): critical difference at the 5 % probability level; NS: non-significant; STBN: soil test-based nutrient recommendation; FYM: farmyard manure; VC: vermicompost; NU: nano urea; ZnSO₄: zinc sulphate.

registered 28 % maximum grain yield of 5804 kg ha⁻¹ (pooled value) and 19 % more straw yield of 6598 kg ha⁻¹ (pooled value) but was statistically at par with application of 75 % STBN + 25 % FYM + ZnSO₄ (T₈), 75 % STBN + NU + ZnSO₄ (T₁₀), 50 % STBN + 25 % VC + NU + ZnSO₄ (T₁₂), 50 % STBN + 25 % FYM + NU + ZnSO₄ (T₁₁) and 100 % STBN + ZnSO₄ (T₇) to rice plants (Table 2). Minimum grain and straw yield of rice was recorded in 100 % STBN (T₁) with pooled value of 4544 kg ha⁻¹ and 5551 kg ha⁻¹ respectively.

Effect of integrated nutrient management on nutrient uptake of rice

Data on pooled analysis revealed maximum N uptake (74.61) in grains with the application of 75 % STBN + 25 % VC + ZnSO₄ (T₉) which was statistically at par with 75 % STBN + 25 % FYM + ZnSO₄ (T₈), 75 % STBN + NU + ZnSO₄ (T₁₀), 50 % STBN + 25 % VC + NU + ZnSO₄ (T₁₂), 50 % STBN + 25 % FYM + NU + ZnSO₄ (T₁₁) and 100 % STBN + ZnSO₄ (T₇) to rice plants (Table 3). However, the N uptake by straw was not significantly influenced by various treatments in both years of study. Zinc is necessary for protein synthesis, photosynthesis, integrity of cellular membranes and increasing nutrition in grains (9). Data on pooled analysis revealed maximum P uptake (17.21 kg ha⁻¹, 5.27 kg ha⁻¹ respectively), K uptake (26.3 kg ha⁻¹ and 101.10 kg ha⁻¹ respectively) and S uptake (4.58 kg ha⁻¹ and 6.42 kg ha⁻¹ respectively) in grains and straw with the application of 75 % STBN + 25 % VC + ZnSO₄ (T₉) which were statistically at par with 75 % STBN + 25 % FYM + ZnSO₄ (T₈), 75 % STBN + NU + ZnSO₄ (T₁₀), 50 % STBN + 25 % VC + NU + ZnSO₄ (T₁₂), 50 % STBN + 25 % FYM + NU + ZnSO₄ (T₁₁) and 100 % STBN + ZnSO₄ (T₇) (Table 3). This might be attributed to better yield, higher P use efficiency and availability of P in presence of FYM. Data on pooled analysis revealed maximum Zn uptake in grains and straw with the application of 75 % STBN + 25 % VC + ZnSO₄ (T₉) with pooled values of 170.86 g ha⁻¹ and 265.42 g ha⁻¹, respectively which was statistically at par with 75 % STBN + 25 % FYM + ZnSO₄ (T₈), 75 % STBN + NU + ZnSO₄ (T₁₀), 50 % STBN + 25 % VC + NU + ZnSO₄ (T₁₂), 50 % STBN + 25 % FYM + NU + ZnSO₄ (T₁₁) and 100 % STBN + ZnSO₄ (T₇).

Discussion

Effect of integrated nutrient management on growth attributes of rice

Zinc application improves plant growth by enhancing physiological functions, nutrient uptake and biomass production (10). Its integrated use with organic manures and inorganic fertilisers sustains soil fertility and corrects Zn deficiency under intensive rice cultivation (11). The current results unequivocally show that physiological growth parameters like LAI, chlorophyll index (SPAD) and Green Seeker readings were greatly enhanced by integrated nutrient management practices that combined inorganic fertilisers (STBN), organic sources (VC/FYM/neem-based inputs) and micronutrient supplementation (ZnSO₄). The combined application of zinc (Zn) and vermicompost enhanced nutrient uptake through their complementary effects on soil and plant processes. Vermicompost improved soil structure, microbial activity and nutrient availability while reducing Zn fixation through the release of organic acids and humic substances. Zinc promoted root growth, enzyme activation and auxin synthesis, leading to greater root growth and nutrient absorption. Together, these synergistic effects improved NUE, resulting in enhanced nutrient uptake and better crop growth and productivity. These results prove that balanced nutrient management is essential for improving crop growth, productivity and soil health. They also closely align with previous reports and other current studies (12). According to studies, applying organic manures and mineral fertilisers together enhances source-sink connections, increasing LAI and biomass accumulation because of enhanced soil physical conditions and prolonged nutrient availability (13). There have also been reports of a high positive correlation between nitrogen treatment and SPAD levels in rice, which suggests greater chlorophyll production and increased plant vigour (14). Furthermore, optimised fertiliser supply improves N absorption and utilisation efficiency, resulting in increased chlorophyll content and improved crop development, according to SPAD-based N management research (15). Zinc fertilisation

Table 3. Nutrient uptake by grain and straw as influenced by integrated nutrient management (INM) (pooled data of 2022–23 and 2023–24)

Treatments	N uptake		P uptake		K uptake		S uptake		Zn uptake	
	Grain	Straw	Grain	Straw	Grain	Straw	Grain	Straw	Grain	Straw
T ₁ 100 % STBN	54.80	27.68	12.88	9.46	19.80	74.52	2.62	4.01	76.86	132.63
T ₂ 75 % STBN + 25 % FYM	57.94	28.16	13.99	9.28	21.70	85.02	2.79	4.85	122.79	194.50
T ₃ 75 % STBN + 25 % VC	60.22	28.36	14.11	9.34	22.00	87.28	2.82	5.16	125.57	197.84
T ₄ 75 % STBN + NU	55.02	27.78	13.18	9.18	19.95	78.28	2.64	4.28	110.60	171.41
T ₅ 50 % STBN + 25 % FYM + NU	56.14	27.85	13.58	9.22	20.65	81.46	2.65	4.53	114.80	164.93
T ₆ 50 % STBN + 25 % VC + NU	56.92	27.97	13.72	9.26	21.20	83.42	2.75	4.73	118.75	191.68
T ₇ 100 % STBN + ZnSO ₄	72.40	28.88	15.88	10.21	24.17	93.68	4.21	5.89	159.24	248.45
T ₈ 75 % STBN + 25 % FYM + ZnSO ₄	74.25	30.62	16.85	10.26	25.70	98.61	4.46	6.39	168.59	261.22
T ₉ 75 % STBN + 25 % VC + ZnSO ₄	74.61	30.98	17.21	10.27	26.30	100.10	4.58	6.42	170.86	265.42
T ₁₀ 75 % STBN + NU + ZnSO ₄	73.29	29.66	16.58	10.25	24.85	97.14	4.36	6.11	164.44	257.17
T ₁₁ 50 % STBN + 25 % FYM + NU + ZnSO ₄	72.65	29.26	15.99	10.23	24.33	94.90	4.24	5.92	160.84	250.93
T ₁₂ 50 % STBN + 25 % VC + NU + ZnSO ₄	72.77	29.49	16.18	10.24	24.60	95.80	4.33	6.03	162.00	254.27
SEm (±)	1.250	1.190	0.729	0.226	0.766	2.803	0.329	0.224	4.059	8.905
CD (0.05)	3.56	NS	2.08	0.65	2.18	7.99	0.94	0.64	11.57	25.38

Values are means of three replications. Pooled data represent the average of 2022–23 and 2023–24. Nutrient uptake is expressed in kg ha⁻¹ for N: nitrogen; P: phosphorus; K: potassium; S: sulphur and g ha⁻¹ for Zn: zinc. SEm (±): standard error of the mean; CD (0.05): critical difference at the 5 % probability level; NS: non-significant; STBN: soil test-based nutrient recommendation; FYM: farmyard manure; VC: vermicompost; NU: nano urea; ZnSO₄: zinc sulphate.

dramatically increases biomass output, chlorophyll content and NUE in rice, according to recent research, improving crop performance overall (16). Furthermore, because of increased soil microbial activity and nutrient availability, integrated nutrition management combined with Zn supplementation has been shown to improve growth characteristics and nutrient absorption (17). Better N status and photosynthetic activity are indicated by an enhanced normalised difference vegetation index (NDVI) under integrated nutrient delivery, which eventually leads to increased production. Under balanced fertilisation and precision nutrient management techniques, comparable gains in spectral indicators and crop growth have been documented (18). Vermicompost increases soil microbial activity and nutrient availability and zinc is essential for enzyme activation and chlorophyll production, both of which raise the content of chlorophyll in leaves (19). This demonstrates how integrated nutrient management may lower reliance on chemical fertilisers while maintaining crop output.

Effect of integrated nutrient management on yield attributes and yield of rice

Recent research has shown similar outcomes, with higher nutrient intake and better assimilate partitioning towards reproductive organs leading to a considerable increase in panicle number, grain number and panicle weight (20). Because of continuous nutrient release and enhanced soil fertility, integrated nutrient management treatments have been demonstrated to offer noticeably superior yield characteristics and productivity when compared to solitary chemical fertilisation (21). Moreover, Zn has been identified as a critical micronutrient limiting rice productivity in eastern India and its application significantly improves yield performance under field conditions (22). Similar results have been documented in which the combined application of Zn fertiliser and organic manures produced the highest grain and straw yield because of increased biomass production and higher NUE (23). Applying VC to various field crops decreased the need for chemical fertilisers without lowering crop yields (24). In comparison to rice cultivated as a single crop or as a component crop in a cropping sequence of 3 or 4 crops, the integration of several plant nutrient sources (such as FYM, VC, crop residues, etc.) has a favourable impact on all yield parameters (25). The maximum rice grain yield and dry matter output may have resulted from the rice crop's nutrient availability being nicely synchronised by the integrated use of organic and inorganic fertiliser sources (26). In both years of the experiment, there was no discernible variation in the HI of rice plants under different integrated nutrition management regimens.

Effect of integrated nutrient management on nutrient uptake of rice

The increased nitrogen intake in grain under integrated treatments is consistent with studies that VC, which improves mineralisation and prolongs nutrient release, greatly boosts N uptake in rice when inorganic fertilisers are substituted (27). In a similar vein, combining organic and inorganic nutrient sources improves soil biological and electrochemical characteristics, which increases nitrogen availability and crop absorption (28). The present study found that better soil chemical characteristics and nutrient dynamics were responsible for considerably increased P and K absorption under integrated nutrient management (29). Similarly, increased K uptake under integrated treatments is associated with improved soil cation exchange capacity and better root proliferation. Recent research on rice nutrient dynamics has shown a similar increase in nutrient

absorption because of applying both organic and inorganic sources (30). Under ZnSO₄-applied treatments, zinc absorption in straw and grain was considerably increased, especially in T₉. This is in line with observations indicating zinc fertilisation boosts enzymatic activity and physiological processes, which greatly increases Zn absorption, nutrient-use efficiency and total rice yield (16). Moreover, because of enhanced root development and nutrient absorption, recent research has demonstrated that integrated nutrition management in conjunction with Zn supplementation dramatically boosts macro- and micronutrient intake in rice (23). This implies that crop performance is not compromised by integrated nutrient management, which increases NUE. Integrated nutrient management has been shown to improve soil fertility and nitrogen absorption while lowering reliance on chemical fertilisers (31).

Correlation studies

Correlation analysis depicted that correlation coefficient values between grain yield with most of the growth and yield attributes was more than 0.9 which indicates grain yield has a strong positive correlation with leaf area index, chlorophyll index readings, green seeker readings, straw yield. Correlation coefficient value between grain yield and harvest index was less than 0.9 which indicates less strong correlation compared to other treatments.

Regression studies

Linear regression analysis was performed to examine the relationship between grain yield and major growth and yield attributes. The relationship between grain yield (Y) and LAI (X) was best described by the equation $y = 3288.4x - 10171$ ($R^2 = 0.9467$, $p < 0.005$), indicating that approximately 94.67 % of the variation in grain yield was explained by LAI (Fig. 1). Similarly, grain yield exhibited a significant positive relationship with chlorophyll index, represented by the equation $y = 107.25x - 10.803$ ($R^2 = 0.9839$, $p < 0.005$) (Fig. 2). The regression between grain yield and GreenSeeker readings was also highly significant ($y = 5188.2x + 1531$, $R^2 = 0.9754$, $p < 0.005$) (Fig. 3). Straw yield showed the strongest association with grain yield and was described by the equation $y = 1.1236x - 1593.6$ ($R^2 = 0.9883$, $p < 0.005$) (Fig. 4). The HI also exhibited a positive linear relationship with grain yield ($y = 743.02x - 29117$), although the relationship was comparatively weaker ($R^2 = 0.8529$, $p < 0.005$) (Fig. 5). Residual diagnostic analysis indicated that residuals were randomly distributed around zero with no discernible pattern, while normal probability plots confirmed approximate normality of residuals. Homoscedasticity of residual variance was also observed, indicating that the assumptions of linear regression were satisfactorily met.

Conclusion

The present study demonstrated that integrated nutrient management (INM) improved rice growth, yield and nutrient uptake under irrigated conditions. The most effective treatment was a combination of 75 % soil test-based nitrogen (STBN) and 25 % nitrogen (N) provided by vermicompost (VC) and zinc sulphate (ZnSO₄). Plant height, tiller number, dry matter accumulation, leaf area index (LAI) and chlorophyll content were all considerably improved by this combination, suggesting increased physiological activity and greater crop vigour. Higher grain and straw yields resulted from a significant increase in yield parameters such as panicle number, filled grains per panicle and test weight. The synergistic interaction of organic and inorganic nutrient sources may

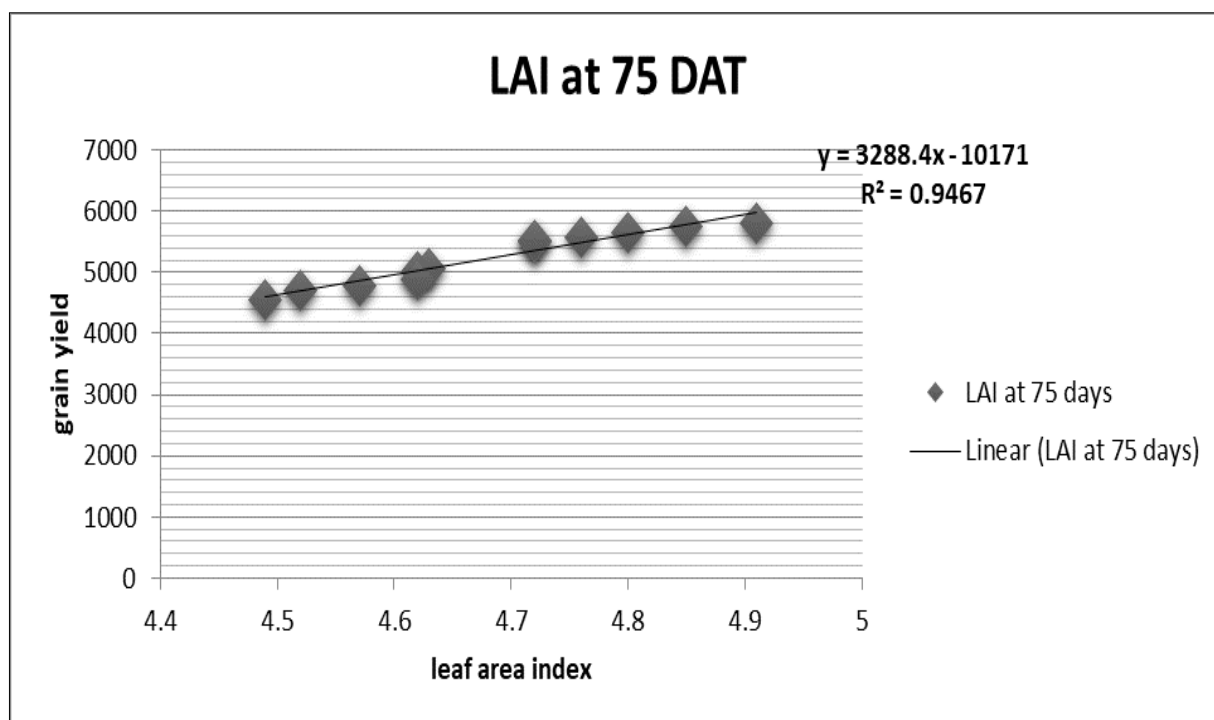


Fig. 1. Regression graph between leaf area index (LAI) at 75 days after transplanting (DAT) and grain yield.

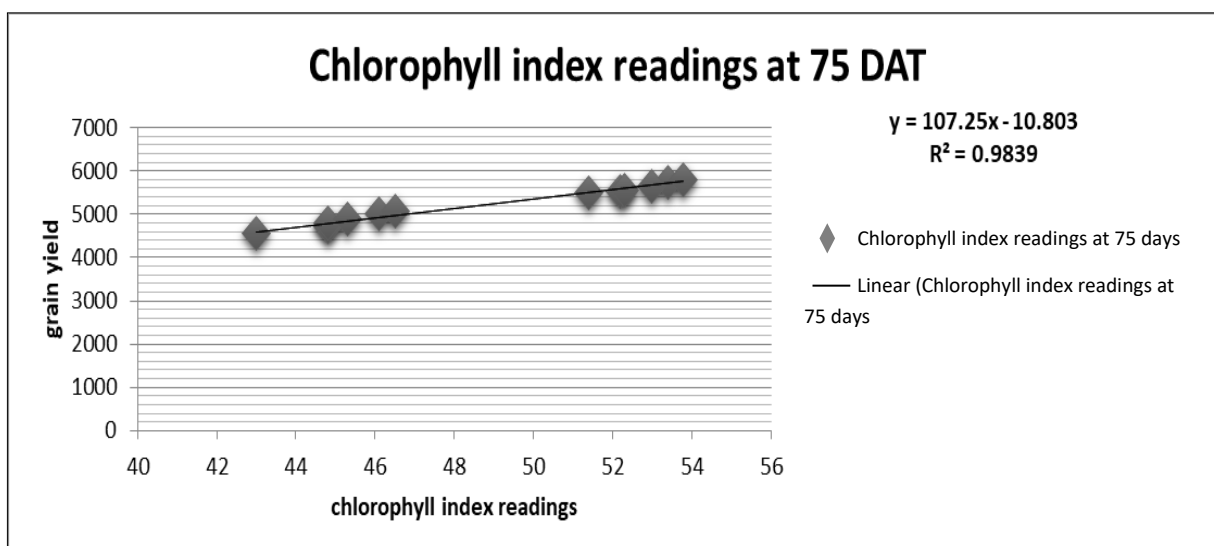


Fig. 2. Regression graph between chlorophyll index readings at 75 days after transplanting (DAT) and grain yield.

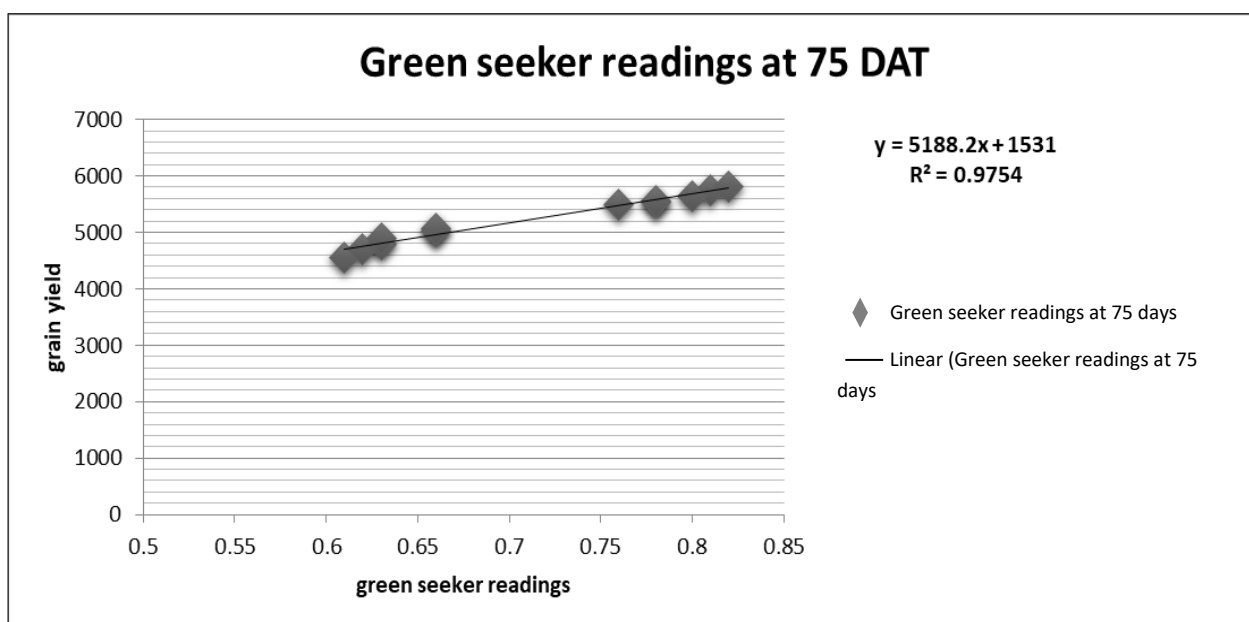


Fig. 3. Regression graph between green seeker readings at 75 DAT and grain yield.

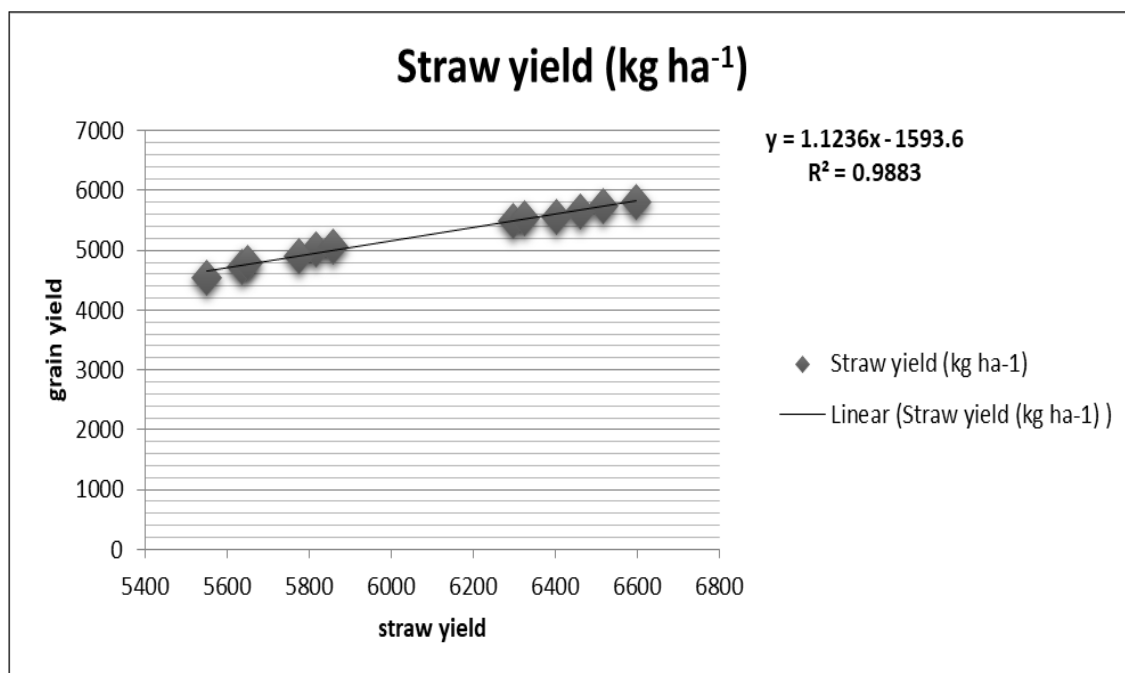


Fig. 4. Regression graph between straw yield and grain yield.

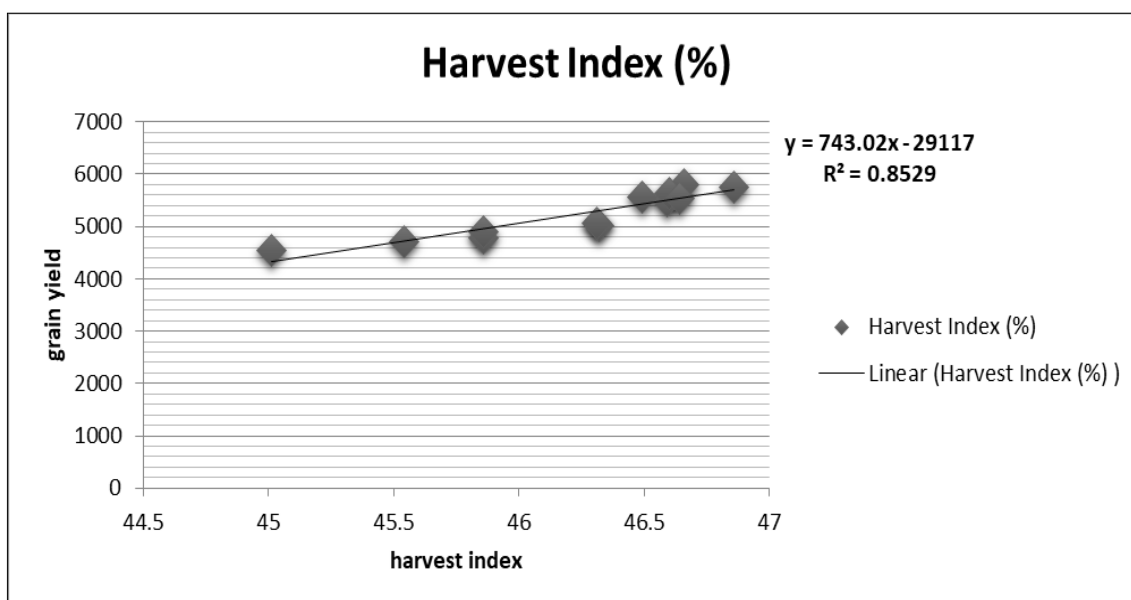


Fig. 5. Regression graph between harvest index and grain yield.

have contributed to improved nutrient availability and crop performance. By enhancing enzyme activities, chlorophyll production and nutrient utilisation efficiency, zinc (Zn) supplementation further improved crop performance. Furthermore, increased absorption of micronutrients (Zn) and macronutrients [N, phosphorus (P), potassium (K) and sulphur (S)] under integrated treatments indicates better nutritional availability and effective absorption. The relevance of balanced nutrition in optimising productivity is confirmed by the substantial positive association between grain yield and growth as well as yield characteristics. Despite the promising results, further research is needed to evaluate the long-term sustainability of INM across diverse agro-climatic conditions and soil types. Future studies should optimise the use of nano fertilisers, organic sources and biofertilisers while assessing soil health, environmental impacts, economic feasibility and farmer adoption to develop sustainable nutrient management strategies for rice production.

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Authors' contributions

SSN contributed to conceptualisation, investigation, data analysis and preparation of the original draft. ST contributed to investigation, data curation and manuscript review. AKP contributed to conceptualisation, methodology, supervision and manuscript editing. RKN contributed to experimental planning and data interpretation. AB, UG and GP contributed to investigation and data curation. AKS and DDP contributed to data analysis, manuscript editing and visualisation. ACD contributed to laboratory analysis and data validation. SKS contributed to technical support and

methodology. PMB contributed to formal analysis and validation. AG and GN contributed to technical guidance and manuscript review. GPP contributed to supervision, project administration and final manuscript review. All authors have read and approved the final manuscript.

Compliance with ethical standards

Conflict of interest: Authors do not have any conflict of interest to declare.

Ethical issues: None

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